5.1

1 (1)

$$y = \tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta} = \frac{2x}{1-x^2}$$

$$\therefore \quad (1-x^2)\,y = 2x$$

2 (1)

$$x = \frac{1}{t-1}, \quad y = \frac{1}{t^2-1}$$

$$\frac{dx}{dt} = \frac{-1}{(t-1)^2}, \quad \frac{dy}{dt} = \frac{-2t}{(t^2-1)^2}$$

$$\therefore \quad \frac{dy}{dx} = \frac{-2t}{(t^2-1)^2} \cdot \left(-\frac{(t-1)^2}{1}\right) = \frac{2t}{(t+1)^2}$$

$$\therefore \quad \frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt} = \frac{\dfrac{2(t+1)^2 - 2t\cdot 2(t+1)}{(t+1)^4}}{-\dfrac{1}{(t-1)^2}} = \frac{2(t+1)-4t}{(t+1)^3}\cdot(-(t-1)^2)$$

$$= -\frac{(-2t+2)(t-1)^2}{(t+1)^3} = \frac{2(t-1)^3}{(t+1)^3}$$

5

$$t=\frac{5}{4}\pi \quad \cdots \quad x(t) = 3\sin t - 4, \quad y(t) = 5 + 2\cos t$$

$$\frac{dx}{dt} = 3\cos t, \quad \frac{dy}{dt} = -2\sin t$$

$$\frac{dy}{dx} = \frac{-2\sin t}{3\cos t} = -\frac{2}{3}\tan t$$

$$\left.\frac{dy}{dx}\right|_{t=\frac{5}{4}\pi} = -\frac{2}{3}\tan\frac{5}{4}\pi = -\frac{2}{3}\cdot 1 = -\frac{2}{3}$$

$$x\left(\frac{5}{4}\pi\right) = 3\cdot\left(-\frac{\sqrt{2}}{2}\right) - 4 = -\frac{3}{2}\sqrt{2} - 4$$

$$y\left(\frac{5}{4}\pi\right) = 5 - 2\cdot\frac{\sqrt{2}}{2} = 5 - \sqrt{2}$$

접방 : $y = -\frac{2}{3}\left(x + \frac{3}{2}\sqrt{2} + 4\right) + 5 - \sqrt{2}$

법방 : $y = \frac{3}{2}\left(x + \frac{3}{2}\sqrt{2} + 4\right) + 5 - \sqrt{2}$

7(2)

$$y = tx$$

$$x^2 + 2tx^2 + 4t^2x^2 = 8x$$

$$x + 2tx + 4t^2x = 8$$

$$\therefore \quad x = \frac{8}{1+2t+4t^2} \,,,$$

$$\therefore \quad y = \frac{8t}{1+2t+4t^2}$$